

Topic 17: Inheritance

17.1 Chromosomes, Genes, and DNA

Term	Definition	Key Facts
Inheritance	The transmission of genetic information from generation to generation.	This is the basis of all genetic study.
Chromosome	A thread-like structure of DNA, found in the nucleus, that carries genetic information in the form of genes .	Humans have 46 chromosomes (23 pairs) in most body cells.
Gene	A short length of DNA found on a chromosome that codes for a specific protein .	Genes control characteristics by coding for structural proteins, enzymes, or hormones.
Allele	A version or form of a gene .	An individual inherits two alleles for each gene (one from each parent).
Diploid Nuclei	Nuclei with two sets of chromosomes (2n).	Found in most body cells (somatic cells). In humans, $2n=46$.
Haploid Nuclei	Nuclei with one set of unpaired chromosomes (n).	Found in gametes (sex cells: sperm and egg). In humans, $n=23$.

17.2 Cell Division

Mitosis

Mitosis is a type of nuclear division that gives rise to two genetically identical cells (clones).

Feature	Description	Importance
Result	Two diploid (2n) daughter cells.	Used for growth, repair of damaged tissues, replacement of cells, and asexual reproduction.
Process	Chromosomes double and are pulled apart, ensuring each daughter cell receives an identical set of chromosomes.	Maintaining the same genetic information in all somatic cells.

Meiosis

Meiosis is reduction division in which the chromosome number is halved from diploid to haploid resulting in genetically different cells (gametes).

Feature	Description	Importance
Result	Four haploid (n) daughter cells.	Used for the production of gametes (sperm and egg).
Process	Involves two divisions, reducing the chromosome number by half (reduction division) and introducing genetic variation through recombination.	Ensures that the zygote formed at fertilisation has the correct diploid number of chromosomes.

17.4 Monohybrid Inheritance

This is the inheritance of characteristics controlled by a single gene [6].

Term	Definition	Example
Phenotype	The observable characteristics of an organism (what it looks like).	Tall, Short, Brown Eyes.
Genotype	The combination of alleles that control a characteristic.	TT, Tt, tt.
Dominant Allele	An allele that is always expressed in the phenotype, even if only one copy is present (e.g., T).	TT (homozygous dominant) and Tt (heterozygous) both show the dominant phenotype.
Recessive Allele	An allele that is only expressed in the phenotype if two copies are present (e.g., t).	tt (homozygous recessive) shows the recessive phenotype.
Homozygous	Having two identical alleles for a gene (e.g., TT or tt).	Also known as 'pure breeding'.
Heterozygous	Having two different alleles for a gene (e.g., Tt).	Also known as a 'hybrid'.

Punnett Squares: A diagram used to show the possible combinations of alleles that could be produced in the offspring and to determine the ratio of genotypes and phenotypes.

Codominance

Codominance occurs when both alleles in a heterozygous organism are expressed in the phenotype [7].

- Example: ABO Blood Group System.
- Alleles I^A and I^B are codominant.
- Allele I^O is recessive to both I^A and I^B .
- A person with genotype $I^A I^B$ will have blood group AB (both A and B antigens are expressed).

Genotype	Phenotype (Blood Group)
$I^A I^A$ or $I^A I^O$	A
$I^B I^B$ or $I^B I^O$	B
$I^A I^B$	AB (Codominant)
$I^O I^O$	O

Sex-Linked Characteristics

These are characteristics controlled by genes located on the sex chromosomes (usually the X chromosome) [8].

- Sex Determination: Females have XX chromosomes; males have XY chromosomes.
- Mechanism: The Y chromosome is much smaller and carries very few genes. Therefore, males only have one copy of the genes found on the X chromosome.
- Result: Males are more likely to express recessive characteristics (like red-green colour blindness or haemophilia) controlled by genes on the X chromosome because they do not have a second X chromosome to mask the recessive allele.
- Carrier: A female who is heterozygous for a sex-linked recessive condition (e.g., $XHxh$ for haemophilia) is a carrier. She does not have the condition but can pass the recessive allele to her offspring.

PLZ note that this file might be incomplete

We are working on it